

Global Valuation

Cross-Country Price-Earnings Multiples

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Data and Sample

Firm-Level Regressions

Country Effects Are Large and Persistent

What Explains Residual Country Effects?

Summary and Next Steps

Data and Sample

Data Sources

Source	Size	Rows	Cols	Description
compustat_global	2.3 GB	981K	468	Global annual fundamentals (122 countries)
compustat_america	0.8 GB	254K	984	North America fundamentals
returns_global	1.9 GB	10.4M	25	Global monthly prices (from 2007)
IBES consensus	WRDS	3.1M	—	Analyst EPS forecasts (FY1, FY2, LTG)

- **Panel:** 743K firm-years, 81 countries (US from 1986; non-US: 65+ countries from 2007, ≤ 7 before). Forward P/E: 169K firm-years, 76 countries.
- **Market cap (Global):** $cshoi \times prccm$ using $prirow$ (primary security)
- **IBES link:** US via ICLINK/CCM; International via `comp.g_security.ibtic`

Variable Definitions

Gordon model motivation:

$$P/E = \frac{1}{r-g} \quad \Rightarrow \quad \log(P/E) = -\log(r-g) \quad (\text{constant growth})$$

Forward P/E uses analyst EPS forecasts $\Rightarrow$ directly captures the **discount-rate channel**.

Forward P/E (FY1) Market Cap / (IBES median FY1 EPS $\times$ shares), positive forecasts

Forward P/E (FY2) Same with two-year-ahead consensus

LTG IBES long-term growth forecast (3–5 year EPS growth, %)

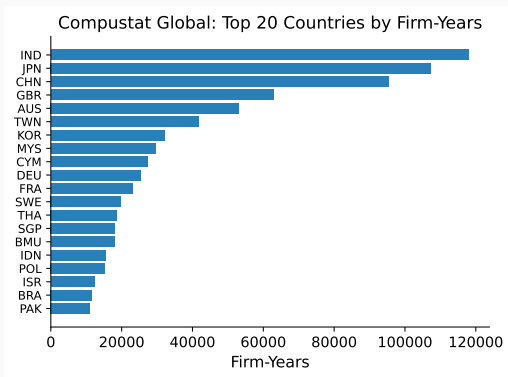
R&D / Assets Firm-level intangible investment intensity

IMF growth WEO 5-year-ahead GDP growth forecast (country-year, from 2008)

Real rate World Bank real interest rate (country-year)

All ratios winsorized at 1st/99th percentiles. IBES consensus requires ≥ 3 analysts. BPN $\div 100$ for UK.

Coverage and Descriptive Statistics



Variable	N	Median	P10	P90
log(P/E)	532K	2.79	1.76	4.32
ROA	743K	0.02	-0.15	0.11
Leverage	743K	0.49	0.14	0.85

Top 5 by obs: USA (188K), JPN (60K), CHN (58K), IND (57K), TWN (31K).

Data availability:

Trailing P/E: 540K obs (1986–2024)

Forward P/E FY1: 169K obs (2000–2024)

IBES LTG: 51K obs (2000–2024)

IMF WEO growth: from 2008 only

Non-US market cap: ≤ 7 countries pre-2007; 65+ countries from 2007 (`returns_global`).

Firm-Level Regressions

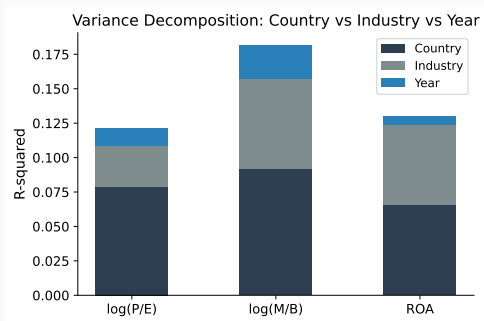
Panel Regression: Forward P/E with Progressive Controls

Dep. variable: $\log(\text{Forward P/E FY1})_{i,t}$ (positive analyst forecasts)

	(1) Base	(2) + R&D	(3) + LTG	(4) + Macro
Leverage	−0.67*** (0.11)	−0.65*** (0.11)	−0.82*** (0.14)	−0.93*** (0.17)
$\log(\text{Assets})$	+0.00 (0.03)	+0.01 (0.03)	+0.04 (0.04)	+0.06 (0.06)
Lag ΔE	−0.01***	−0.01***	−0.01***	−0.01
R&D / Assets		+1.73*** (0.44)	+2.44*** (0.27)	+3.49*** (0.42)
IBES LTG (%)			+0.013*** (0.002)	+0.012*** (0.002)
Real int. rate				+0.004 (0.01)
IMF growth 5y				+0.083* (0.05)
N / R^2	150K / 0.13	150K / 0.14	44K / 0.15	21K / 0.16
$\sigma(\hat{\gamma}_c) / \text{Ctry}$	0.56 / 76	0.55 / 76	0.71 / 58	0.56 / 33

SE clustered by country. All specs: Country + Industry + Year FE. LTG and R&D strongly priced. Real rate **null**; IMF growth marginal. Country effects **persist** ($\sigma = 0.71$).

Variance Decomposition: Country vs Industry vs Year



Shapley R^2 decomposition of $\log(P/E)$:

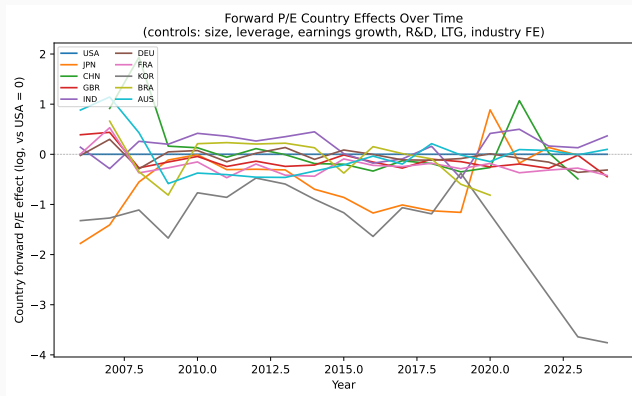
Factor	Share of R^2
Country	64.6%
Industry	24.8%
Year	10.6%

Country alone explains nearly two-thirds of the variation in P/E .

Extends Heston–Rouwenhorst (1994) from *returns* to *valuation levels*.

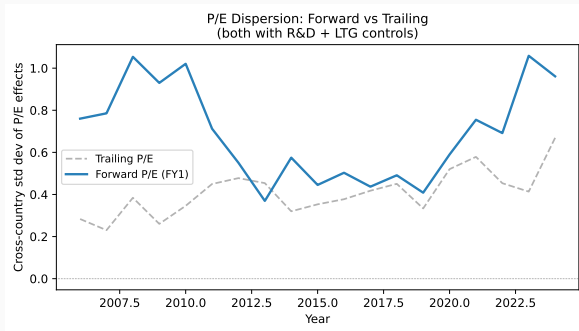
Country Effects Are Large and Persistent

Country Forward P/E Effects Over Time (vs USA)



Year-by-year regressions with size, leverage, earnings growth, R&D, analyst LTG, industry FE. USA = 0. Japan and Korea: persistent discount. India rising. China volatile but near zero after LTG control.

Forward P/E Dispersion Is Diverging



Forward P/E dispersion:

$$\sigma(\hat{\gamma}_c) \approx 0.4-1.0$$

Forward P/E shows **larger** cross-country dispersion than trailing P/E throughout.

Removing expected earnings differences *widens* country gaps.

⇒ Country P/E gaps reflect **discount rates**, not earnings expectations.

What Explains Residual Country Effects?

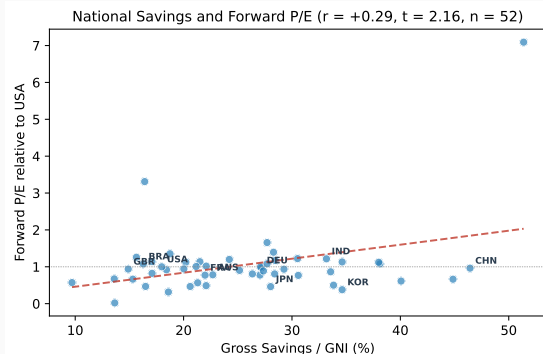
Cross-Country Regressions: Forward P/E Effects

Dep. variable: country forward P/E effect $\hat{\gamma}_c$ (log scale, USA = 0, from pooled regression with all firm-level controls)

	(1)	(2)	(3)	(4)	(5)
Savings / GNI	+0.024** (2.16)		+0.022* (1.91)	+0.028** (2.48)	
Mkt Cap / GDP		+0.001 (1.13)	−0.000 (−0.06)		
Credit / GDP		−0.005* (−1.96)		−0.004* (−1.79)	
Anti-Self-Dealing					−0.099 (−0.38)
R^2	0.09	0.08	0.08	0.15	0.00
N	52	46	48	49	44

OLS t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Savings/GNI: modest positive predictor ($r = +0.29$). Financial development variables: null. Residual country effects remain **largely unexplained**.

National Savings and Forward P/E



Gross savings / GNI vs forward P/E
country effect:

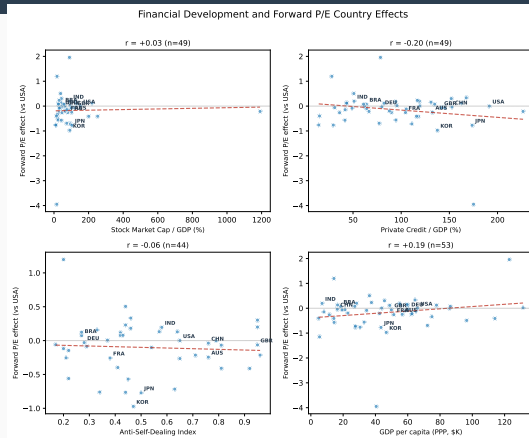
$$r = +0.29, \quad t = 2.16^{**} \quad (n = 52)$$

After controlling for all firm-level variables
(R&D, LTG, leverage, size, earnings growth)
+ industry + year FE.

Demand-side channel: more capital seeking
equity returns $\Rightarrow$ higher price multiples.

But $R^2 = 0.09$ — savings explains <10% of
residual country effects.

Financial Development Does Not Predict Forward P/E



Market cap/GDP ($r = +0.03$), credit/GDP ($r = -0.20$), anti-self-dealing ($r = -0.06$), GDP per capita ($r = +0.19$): all near zero or wrong-signed. La Porta et al. (2002) variables do not explain valuation levels after firm-level controls.

Summary and Next Steps

Stylized Facts

1. **Country dominates P/E** — 64.6% of R^2 (Shapley), after controlling for size, leverage, earnings growth, industry.
2. **Forward P/E widens country gaps** — Removing expected earnings increases dispersion. Country effects reflect **discount rates**, not earnings.
3. **Large and persistent** — $\sigma(\hat{\gamma}_c) \approx 0.7$ even after R&D, LTG, macro controls. Japan trades at $0.46\times$ USA forward P/E. India at $1.22\times$.
4. **Real interest rate: null** — Cross-country real rates do not predict forward P/E ($t = 0.49$). IMF growth forecasts marginal ($t = 1.72$). Consistent with the “stock-yield disconnect” (Gormsen–Lazarus 2026).
5. **Savings: modest predictor** — Savings/GNI predicts forward P/E ($r = +0.29$), but financial development variables (investor protection, market depth) are null.

Research Questions

1. **What drives the forward P/E discount-rate gap?** — After controlling for firm-level growth, R&D, and risk proxies, large country effects remain. What are we missing?
2. **Can the savings channel be formalized?** — Savings/GNI is the best cross-sectional predictor. A model where savings rates affect discount rates?
3. **Do low-P/E countries earn higher returns?** — If so, valuation gaps reflect risk premia. Expected-returns test using Fama–French international factors.
4. **Pure discounting decomposition** — Gormsen–Lazarus (2026): can their “pure discounting” component explain cross-country forward P/E?
5. **Why is Japan so cheap?** — Forward P/E at $0.46\times$ USA even after analyst growth controls. Korea even lower ($0.38\times$). Cultural, governance, or measurement?

Related Literature

Country vs industry effects:

- Heston–Rouwenhorst (1994) — country dominates returns. We extend to *levels*.
- Griffin–Karolyi (1998) — industry effects vary across sectors.

Institutions and investor protection:

- La Porta et al. (2002) — investor protection raises Tobin's Q. Weak for forward P/E.
- Hail–Leuz (2006) — institutions reduce cost of equity by up to 200 basis points.
- Stulz (2005) — “twin agency problems” prevent equalization.

Savings and asset prices:

- Abel (2003) — baby boomers' saving and asset prices.
- Caballero et al. (2008) — global imbalances and asset price inflation.

Interest rates and equity valuations:

- Gormsen–Lazarus (2026) — “stock-yield disconnect”: only “pure discounting” passes through to equities.

Appendix

Full List of Tested Predictors

Variable	r (trailing)	t -stat	N
IBES Long-Term Growth	+0.50***	3.48	39
R&D / Assets	+0.48***	4.73	76
Savings / GNI	+0.34***	3.13	68
ROA (−0.08 after R&D ctrl)	−0.39	−3.75	76
WHT (n.s. after R&D ctrl)	+0.38**	2.85	51
Lending Rate	−0.33*	−2.00	34
Combined Dividend Tax	+0.25*	1.82	50
Private Credit / GDP	+0.17	1.30	72
CIT Rate	+0.16	1.17	51
IMF GDP growth forecast	+0.13	1.07	73
Anti-Self-Dealing Index	+0.13	0.92	52
CapEx / Assets	+0.03	0.24	78
GDP per capita (PPP)	+0.01	0.09	73
Real Interest Rate	−0.03	−0.15	34
Realized earnings growth	−0.00	−0.04	80

Sorted by $|r|$. Based on *trailing* P/E country effects (base regression, not forward P/E). ROA and WHT absorbed by R&D.

IBES Merge and Coverage Details

Linking strategy:

- US: ICLINK (wrdsapps.ibcrsphist) → CCM (crsp.ccmxpf_lnkhist) → gvkey
- International: comp.g_security.ibtic → gvkey (41K direct pairs)
- Combined: 62K unique IBES tickers linked to Compustat

Coverage (forward P/E sample):

Country	FY1 obs	LTG obs	% of trailing sample
USA	84K	34K	45%
JPN	31K	2K	52%
GBR	14K	4K	65%
CHN	4K	0.4K	7%
IND	7K	1K	12%

UK EPS in British Pence (BPN): divided by 100. Consensus requires ≥ 3 analysts.

Market Cap Construction (Detail)

US firms: `mkvalt` (pre-computed market value in USD millions). Available from 1986.

Non-US firms (Compustat Global): no pre-computed market value.

- Compute: $\text{Market Cap} = \text{cshoi}$ (shares outstanding) $\times$ `prccm` (month-end price)
- **Pitfall:** firms have multiple securities across exchanges and currencies

Example — SAP SE (gvkey=103487) in Dec 2022:

`iid=02W` `EUR` 96.39 ← correct

`iid=05W` `HUF` 38,620 ← wrong currency!

`iid=06W` `MXN` 2,009 ← wrong currency!

- **Solution:** use `prirow` field (primary security designation)

All data in local currency (millions). Cross-country *levels* absorbed by country FE.